Project: Worked examples
Title: ISO 286 limits and fits
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1. Scope and method

Hole and shaft limits for a nominal size and tolerance classes, the resulting fit (clearance, transition or interference) and its extreme values, from the ISO 286-1 formulae.

WHAT IT CALCULATES	RELATION
Standard tolerance factor at the geometric mean D of the size step	i = 0.45·D^(1/3) + 0.001·D (µm)
Width of the tolerance band for the grade	ITn = multiplier × i
Fundamental deviations by letter from the published formulae, and hole deviations by the general rule	EI = −es (A–H); ES = −ei + Δ (K–ZC)
Limits of size of the hole and the shaft	limit = nominal + deviation
Fit extremes, the largest and smallest clearance (negative is interference)	c_{max} = ES − ei; c_{min} = EI − es
Rolling-bearing bore mode	a bearing bore is not an ISO 286 hole; it has its own tolerance (ISO 492) lying below nominal, so ES and EI are entered from the bearing record instead of being computed

Basis: Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations[1](§4, §5); Tables of standard tolerance classes and limit deviations (not reproduced)[2].

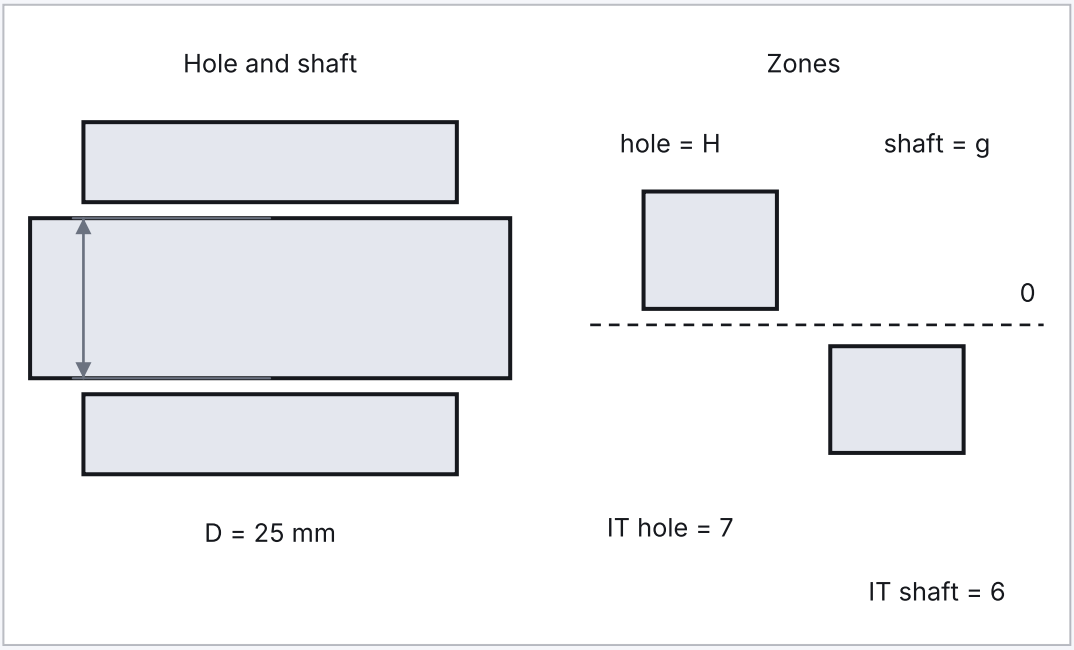


Figure 1 — A hole and its shaft of one nominal size D, and their tolerance zones about the zero line (hole above, shaft below as drawn; the letters place them). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D	Nominal size	25	mm	drawing: DRG-SHAFT-12 rev A	1 … 500 mm
$hole_{kind}$	Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)	ISO 286 hole class		default	any
$hole_{letter}$	Hole fundamental deviation letter	H		spec: DRG-SHAFT-12 note 3: H7/g6	any
$hole_{grade}$	Hole IT grade	7		spec: DRG-SHAFT-12 note 3	5 … 11
$shaft_{letter}$	Shaft fundamental deviation letter	g		spec: DRG-SHAFT-12 note 3	any
$shaft_{grade}$	Shaft IT grade	6		spec: DRG-SHAFT-12 note 3	4 … 11

3. Assumptions

A1. IT values and fundamental deviations are the unrounded results of the ISO 286-1 formulae at the geometric mean of the size step. (The tabulated values apply the standard's own rounding and may differ by about 1 µm for the clearance and transition letters (a to n). For the interference letters p to zc the tables also carry the standard's own increments and sub-steps, so the formula value can be several µm below the tabulated one (p6 at 450 mm: 62.2 against 68 µm), and the sheet warns; enter overrides from a released reference record where the tabulated value governs.)
A2. Hole deviations follow the general rule from the shaft deviations, with Δ = ITn − IT(n−1) where the standard applies it. (Table-specific departures in the standard are not reproduced; use the overrides where they govern.)

4. Working

[7] D_{step} = size step (lower, upper] [1](§4)
= (18, 30] mm
= 23.238 mm
Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D [1](§5)
= D = 23.238 mm
= 1.3074 µm
Standard tolerance unit

[9] IT_{hole} = IT7 [1](§5)
= 16 × i
= 20.918 µm
Hole tolerance

[10] IT_{shaft} = IT6 [1](§5)
= 10 × i
= 13.074 µm
Shaft tolerance

[11] es = fundamental deviation 'g' [1](§5)
= -2.5·D^0.34
= -7.2853 µm
Shaft upper deviation

[12] $ei = es - IT_{shaft}$ [1](§5)
$= -7.2853 - 13.074$
$= -20.359\ \mu m$
Shaft lower deviation
[13] $ES = EI + IT_{hole}$ [1](§5)
$= 0 + 20.918$
$= 20.918\ \mu m$
Hole upper deviation
[14] EI = fundamental deviation 'H' [1](§5)
= -es of the same letter
$= 0\ \mu m$
Hole lower deviation
[15] $D_{hole,max} = D + ES$
$= 25\ mm + 20.918\ \mu m$
$= 25.021\ mm$
Hole maximum
[16] $D_{hole,min} = D + EI$
$= 25\ mm + 0\ \mu m$
$= 25\ mm$
Hole minimum
[17] $d_{shaft,max} = D + es$
$= 25\ mm + -7.2853\ \mu m$
$= 24.993\ mm$
Shaft maximum
[18] $d_{shaft,min} = D + ei$
$= 25\ mm + -20.359\ \mu m$
$= 24.98\ mm$
Shaft minimum
[19] $c_{max} = ES - ei$
$= 20.918\ \mu m - (-20.359\ \mu m)$
$= 41.277\ \mu m$
Maximum clearance (negative: minimum interference)
[20] $c_{min} = EI - es$
$= 0\ \mu m - (-7.2853\ \mu m)$
$= 7.2853\ \mu m$
Minimum clearance (negative: maximum interference)
[21] $i_{max} = es - EI$
$= (-7.2853\ \mu m) - 0\ \mu m$
$= -7.2853\ \mu m$
Maximum interference
[22] $i_{min} = ei - ES$
$= (-20.359\ \mu m) - 20.918\ \mu m$
$= -41.277\ \mu m$
Minimum interference (negative: maximum clearance)
[23] fit = fit class
= H7/g6
= "clearance"
Fit kind: clearance, transition or interference
[24] $c_{mean} = (c_{max} + c_{min}) / 2$
$= (41.277\ \mu m + 7.2853\ \mu m) / 2$
$= 24.281\ \mu m$
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
IT_{hole}	20.918 µm	Hole tolerance
IT_{shaft}	13.074 µm	Shaft tolerance
ES	20.918 µm	Hole upper deviation
EI	0 µm	Hole lower deviation
es	-7.2853 µm	Shaft upper deviation
ei	-20.359 µm	Shaft lower deviation
$D_{hole,max}$	25.021 mm	Hole maximum
$D_{hole,min}$	25 mm	Hole minimum
$d_{shaft,max}$	24.993 mm	Shaft maximum
$d_{shaft,min}$	24.98 mm	Shaft minimum
fit	"clearance"	Fit kind
c_{max}	41.277 µm	Maximum clearance
c_{min}	7.2853 µm	Minimum clearance
i_{max}	-7.2853 µm	Maximum interference
i_{min}	-41.277 µm	Minimum interference
c_{mean}	24.281 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classified	H7/g6	clearance	0	

7. Applicability

WARNING — input.hole_kind: $hole_{kind}$ (Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)) was not entered; the template default 'ISO 286 hole class' is used.

8. Verification

```
Template mech.iso286-fit@1.3.0 Content hash c2553872c4e478c5 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 1: Basis of Tolerances, Deviations and Fits. ISO 286-1:2010. ISO; 2010.
2. International Organization for Standardization. Geometrical Product Specifications (GPS)—ISO Code System for Tolerances on Linear Sizes—Part 2: Tables of Standard Tolerance Classes and Limit Deviations for Holes and Shafts. ISO 286-2:2010. ISO; 2010.